Opportunity Cross-Sensor Diffusion Evaluation

Downstream C-LSTM-A recognition after missing-sensor imputation

What this PDF reports

- Dataset: Opportunity (14 IMU sensors, 4 device groups)
- CrossDiff = diffusion imputation of missing sensors
- Mean-Fill = replace missing sensors with training-set mean waveform
- Baseline (all_real) = complete sensors, no missingness
- Reported as Acc / AF1; MAP and AUC included when defined
- Winner decided by Accuracy (same rule as evaluation script)
- Δ Acc = CrossDiff Acc − Mean Acc

Sensors (14)

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back_acc, back_gyro | rua_acc, rua_gyro, rla_acc, rla_gyro |
lua_acc, lua_gyro, lla_acc, lla_gyro | lshoe_acc, lshoe_gyro, rshoe_acc, rshoe_gyro
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Device groups (4)

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back (2) | right_arm (4) | left_arm (4) | shoes (4)
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Scenarios per label track

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45 scenarios = 1 all_real + 28 single-sensor + 8 device-all + 8 only-*
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Source: eval_outputs/opportunity/<track>_metrics.json
Caution: left-arm tracks are Null-dominated; high Acc with very low AF1 needs careful interpretation.

Summary across all 7 Opportunity label tracks

Track	Acc	AF1	MAP	AUC	#cls	CD Acc>	Mean Acc>	Ties	CD AF1>	JSON CD
locomotion	87.32%	89.16%	0.9246	0.9659	5	9/22	13/22	0/22	11/22	9/22
hl_activity	69.28%	69.28%	0.7389	0.9032	6	7/22	15/22	0/22	11/22	7/22
ll_left_arm	78.72%	8.06%	nan	nan	10	15/22	3/22	4/22	10/22	19/22
ll_left_arm_object	78.48%	3.89%	nan	nan	23	8/22	11/22	3/22	8/22	11/22
ll_right_arm	78.07%	52.09%	0.5438	0.9478	14	9/22	13/22	0/22	11/22	9/22
ll_right_arm_object	70.36%	40.95%	0.4027	0.9304	24	9/22	13/22	0/22	10/22	9/22
ml_both_arms	88.97%	56.02%	0.5569	0.9535	18	8/22	14/22	0/22	8/22	8/22

CD Acc> = CrossDiff Acc strictly greater than Mean. JSON CD = win count saved by eval script (ties may count as CrossDiff).

Locomotion

Baseline all_real: Acc=0.8732 AF1=0.8916 MAP=0.9246 AUC=0.9659 n_classes=5 | CrossDiff Acc wins=9/22 (JSON=

Pattern	Baseline Acc/AF1	CrossDiff Acc/AF1	Mean-Fill Acc/AF1	Winner	Δ Acc
all_real	0.873/0.892	—	—	—	—
back_acc	0.873/0.892	0.821/0.846	0.755/0.776	CrossDiff	+0.066
back_gyro	0.873/0.892	0.848/0.871	0.862/0.883	Mean	-0.013
rua_acc	0.873/0.892	0.864/0.885	0.844/0.870	CrossDiff	+0.020
rua_gyro	0.873/0.892	0.850/0.874	0.866/0.885	Mean	-0.016
rla_acc	0.873/0.892	0.847/0.870	0.843/0.866	CrossDiff	+0.005
rla_gyro	0.873/0.892	0.864/0.884	0.872/0.889	Mean	-0.008
lua_acc	0.873/0.892	0.836/0.856	0.858/0.879	Mean	-0.022
lua_gyro	0.873/0.892	0.858/0.881	0.872/0.891	Mean	-0.014
lla_acc	0.873/0.892	0.854/0.875	0.825/0.851	CrossDiff	+0.029
lla_gyro	0.873/0.892	0.864/0.886	0.869/0.887	Mean	-0.005
lshoe_acc	0.873/0.892	0.841/0.867	0.868/0.889	Mean	-0.027
lshoe_gyro	0.873/0.892	0.839/0.867	0.865/0.886	Mean	-0.026
rshoe_acc	0.873/0.892	0.832/0.848	0.864/0.883	Mean	-0.031
rshoe_gyro	0.873/0.892	0.833/0.862	0.865/0.884	Mean	-0.032
back_all	0.873/0.892	0.814/0.841	0.715/0.733	CrossDiff	+0.099
right_arm_all	0.873/0.892	0.799/0.832	0.747/0.777	CrossDiff	+0.053
left_arm_all	0.873/0.892	0.787/0.813	0.769/0.792	CrossDiff	+0.018
shoes_all	0.873/0.892	0.776/0.799	0.844/0.864	Mean	-0.068
only_back	0.873/0.892	0.491/0.448	0.554/0.436	Mean	-0.063
only_right_arm	0.873/0.892	0.422/0.402	0.432/0.272	Mean	-0.009
only_left_arm	0.873/0.892	0.497/0.502	0.375/0.202	CrossDiff	+0.122
only_shoes	0.873/0.892	0.447/0.441	0.378/0.208	CrossDiff	+0.070

High-level Activity (hl_activity)
Baseline all_real: Acc=0.6928 AF1=0.6928 MAP=0.7389 AUC=0.9032 n_classes=6 | CrossDiff Acc wins=7/22 (JSON=

Pattern	Baseline Acc/AF1	CrossDiff Acc/AF1	Mean-Fill Acc/AF1	Winner	Δ Acc
all_real	0.693/0.693	—	—	—	—
back_acc	0.693/0.693	0.667/0.661	0.656/0.658	CrossDiff	+0.010
back_gyro	0.693/0.693	0.662/0.669	0.682/0.682	Mean	-0.020
rua_acc	0.693/0.693	0.651/0.656	0.666/0.678	Mean	-0.015
rua_gyro	0.693/0.693	0.613/0.627	0.680/0.679	Mean	-0.067
rla_acc	0.693/0.693	0.642/0.639	0.670/0.672	Mean	-0.028
rla_gyro	0.693/0.693	0.664/0.677	0.687/0.676	Mean	-0.023
lua_acc	0.693/0.693	0.676/0.670	0.682/0.683	Mean	-0.006
lua_gyro	0.693/0.693	0.668/0.671	0.686/0.688	Mean	-0.017
lla_acc	0.693/0.693	0.668/0.662	0.569/0.600	CrossDiff	+0.099
lla_gyro	0.693/0.693	0.679/0.677	0.670/0.671	CrossDiff	+0.009
lshoe_acc	0.693/0.693	0.640/0.646	0.677/0.669	Mean	-0.037
lshoe_gyro	0.693/0.693	0.684/0.682	0.664/0.669	CrossDiff	+0.020
rshoe_acc	0.693/0.693	0.646/0.625	0.693/0.692	Mean	-0.047
rshoe_gyro	0.693/0.693	0.663/0.668	0.664/0.668	Mean	-0.001
back_all	0.693/0.693	0.633/0.638	0.629/0.630	CrossDiff	+0.004
right_arm_all	0.693/0.693	0.500/0.514	0.591/0.597	Mean	-0.092
left_arm_all	0.693/0.693	0.622/0.585	0.527/0.558	CrossDiff	+0.095
shoes_all	0.693/0.693	0.541/0.488	0.601/0.575	Mean	-0.060
only_back	0.693/0.693	0.244/0.208	0.284/0.183	Mean	-0.039
only_right_arm	0.693/0.693	0.368/0.321	0.342/0.286	CrossDiff	+0.026
only_left_arm	0.693/0.693	0.215/0.191	0.288/0.173	Mean	-0.073
only_shoes	0.693/0.693	0.262/0.241	0.336/0.380	Mean	-0.074

Left-arm Gestures (ll_left_arm)
Baseline all_real: Acc=0.7872 AF1=0.0806 MAP=nan AUC=nan n_classes=10 | CrossDiff Acc wins=15/22 (JSON=19/2

Pattern	Baseline Acc/AF1	CrossDiff Acc/AF1	Mean-Fill Acc/AF1	Winner	Δ Acc
all_real	0.787/0.081	—	—	—	—
back_acc	0.787/0.081	0.796/0.081	0.786/0.081	CrossDiff	+0.010
back_gyro	0.787/0.081	0.787/0.081	0.787/0.081	Mean	-0.000
rua_acc	0.787/0.081	0.787/0.081	0.787/0.081	CrossDiff	+0.000
rua_gyro	0.787/0.081	0.791/0.080	0.787/0.081	CrossDiff	+0.004
rla_acc	0.787/0.081	0.793/0.081	0.785/0.081	CrossDiff	+0.008
rla_gyro	0.787/0.081	0.795/0.081	0.787/0.081	CrossDiff	+0.008
lua_acc	0.787/0.081	0.787/0.081	0.787/0.081	Tie	+0.000
lua_gyro	0.787/0.081	0.788/0.081	0.787/0.081	CrossDiff	+0.001
lla_acc	0.787/0.081	0.788/0.080	0.797/0.089	Mean	-0.009
lla_gyro	0.787/0.081	0.797/0.081	0.788/0.081	CrossDiff	+0.009
lshoe_acc	0.787/0.081	0.790/0.081	0.787/0.081	CrossDiff	+0.003
lshoe_gyro	0.787/0.081	0.793/0.081	0.788/0.081	CrossDiff	+0.005
rshoe_acc	0.787/0.081	0.789/0.081	0.787/0.081	CrossDiff	+0.001
rshoe_gyro	0.787/0.081	0.790/0.081	0.787/0.081	CrossDiff	+0.002
back_all	0.787/0.081	0.795/0.081	0.786/0.081	CrossDiff	+0.009
right_arm_all	0.787/0.081	0.797/0.089	0.785/0.081	CrossDiff	+0.013
left_arm_all	0.787/0.081	0.797/0.081	0.797/0.089	Mean	-0.000
shoes_all	0.787/0.081	0.797/0.089	0.786/0.081	CrossDiff	+0.011
only_back	0.787/0.081	0.797/0.089	0.797/0.089	Tie	+0.000
only_right_arm	0.787/0.081	0.797/0.089	0.797/0.089	Tie	+0.000
only_left_arm	0.787/0.081	0.797/0.089	0.778/0.081	CrossDiff	+0.020
only_shoes	0.787/0.081	0.797/0.089	0.797/0.089	Tie	+0.000

Left-arm + Object (ll_left_arm_object)
Baseline all_real: Acc=0.7848 AF1=0.0389 MAP=nan AUC=nan n_classes=23 | CrossDiff Acc wins=8/22 (JSON=11/22

Pattern	Baseline Acc/AF1	CrossDiff Acc/AF1	Mean-Fill Acc/AF1	Winner	Δ Acc
all_real	0.785/0.039	—	—	—	—
back_acc	0.785/0.039	0.783/0.039	0.786/0.038	Mean	-0.002
back_gyro	0.785/0.039	0.785/0.039	0.785/0.039	Mean	-0.000
rua_acc	0.785/0.039	0.786/0.038	0.785/0.039	CrossDiff	+0.001
rua_gyro	0.785/0.039	0.786/0.038	0.784/0.039	CrossDiff	+0.002
rla_acc	0.785/0.039	0.785/0.039	0.785/0.038	Mean	-0.001
rla_gyro	0.785/0.039	0.786/0.038	0.785/0.039	CrossDiff	+0.001
lua_acc	0.785/0.039	0.785/0.039	0.785/0.039	CrossDiff	+0.000
lua_gyro	0.785/0.039	0.785/0.039	0.786/0.038	Mean	-0.002
lla_acc	0.785/0.039	0.784/0.039	0.786/0.038	Mean	-0.002
lla_gyro	0.785/0.039	0.785/0.039	0.785/0.039	Mean	-0.001
lshoe_acc	0.785/0.039	0.786/0.038	0.784/0.039	CrossDiff	+0.001
lshoe_gyro	0.785/0.039	0.785/0.039	0.785/0.039	Mean	-0.000
rshoe_acc	0.785/0.039	0.785/0.038	0.785/0.039	Mean	-0.000
rshoe_gyro	0.785/0.039	0.785/0.039	0.785/0.039	Tie	+0.000
back_all	0.785/0.039	0.783/0.039	0.786/0.038	Mean	-0.003
right_arm_all	0.785/0.039	0.786/0.038	0.785/0.038	CrossDiff	+0.002
left_arm_all	0.785/0.039	0.779/0.039	0.786/0.038	Mean	-0.007
shoes_all	0.785/0.039	0.786/0.038	0.785/0.039	CrossDiff	+0.002
only_back	0.785/0.039	0.786/0.038	0.786/0.038	Tie	+0.000
only_right_arm	0.785/0.039	0.786/0.040	0.786/0.038	Mean	-0.000
only_left_arm	0.785/0.039	0.786/0.038	0.786/0.038	CrossDiff	+0.000
only_shoes	0.785/0.039	0.786/0.038	0.786/0.038	Tie	+0.000

Right-arm Gestures (ll_right_arm)
Baseline all_real: Acc=0.7807 AF1=0.5209 MAP=0.5438 AUC=0.9478 n_classes=14 | CrossDiff Acc wins=9/22 (JSON

Pattern	Baseline Acc/AF1	CrossDiff Acc/AF1	Mean-Fill Acc/AF1	Winner	Δ Acc
all_real	0.781/0.521	—	—	—	—
back_acc	0.781/0.521	0.770/0.501	0.762/0.490	CrossDiff	+0.008
back_gyro	0.781/0.521	0.777/0.516	0.767/0.468	CrossDiff	+0.009
rua_acc	0.781/0.521	0.767/0.510	0.752/0.385	CrossDiff	+0.015
rua_gyro	0.781/0.521	0.756/0.416	0.737/0.324	CrossDiff	+0.019
rla_acc	0.781/0.521	0.744/0.467	0.665/0.313	CrossDiff	+0.079
rla_gyro	0.781/0.521	0.702/0.296	0.738/0.430	Mean	-0.036
lua_acc	0.781/0.521	0.751/0.402	0.778/0.518	Mean	-0.027
lua_gyro	0.781/0.521	0.779/0.505	0.775/0.513	CrossDiff	+0.004
lla_acc	0.781/0.521	0.772/0.475	0.774/0.511	Mean	-0.002
lla_gyro	0.781/0.521	0.774/0.500	0.778/0.510	Mean	-0.004
lshoe_acc	0.781/0.521	0.762/0.483	0.778/0.521	Mean	-0.017
lshoe_gyro	0.781/0.521	0.776/0.515	0.780/0.518	Mean	-0.004
rshoe_acc	0.781/0.521	0.779/0.516	0.780/0.516	Mean	-0.000
rshoe_gyro	0.781/0.521	0.778/0.511	0.779/0.519	Mean	-0.001
back_all	0.781/0.521	0.767/0.498	0.753/0.452	CrossDiff	+0.014
right_arm_all	0.781/0.521	0.548/0.106	0.615/0.055	Mean	-0.067
left_arm_all	0.781/0.521	0.697/0.239	0.753/0.472	Mean	-0.056
shoes_all	0.781/0.521	0.741/0.403	0.778/0.512	Mean	-0.037
only_back	0.781/0.521	0.615/0.067	0.615/0.054	CrossDiff	+0.001
only_right_arm	0.781/0.521	0.664/0.106	0.717/0.420	Mean	-0.053
only_left_arm	0.781/0.521	0.617/0.088	0.615/0.054	CrossDiff	+0.002
only_shoes	0.781/0.521	0.599/0.060	0.615/0.054	Mean	-0.016

Right-arm + Object (ll_right_arm_object)
Baseline all_real: Acc=0.7036 AF1=0.4095 MAP=0.4027 AUC=0.9304 n_classes=24 | CrossDiff Acc wins=9/22 (JSON

Pattern	Baseline Acc/AF1	CrossDiff Acc/AF1	Mean-Fill Acc/AF1	Winner	Δ Acc
all_real	0.704/0.409	—	—	—	—
back_acc	0.704/0.409	0.693/0.373	0.684/0.326	CrossDiff	+0.009
back_gyro	0.704/0.409	0.692/0.386	0.692/0.364	CrossDiff	+0.001
rua_acc	0.704/0.409	0.687/0.371	0.685/0.372	CrossDiff	+0.002
rua_gyro	0.704/0.409	0.666/0.306	0.657/0.217	CrossDiff	+0.009
rla_acc	0.704/0.409	0.665/0.289	0.647/0.253	CrossDiff	+0.019
rla_gyro	0.704/0.409	0.663/0.287	0.689/0.360	Mean	-0.026
lua_acc	0.704/0.409	0.661/0.339	0.695/0.401	Mean	-0.034
lua_gyro	0.704/0.409	0.685/0.343	0.699/0.400	Mean	-0.014
lla_acc	0.704/0.409	0.684/0.364	0.680/0.332	CrossDiff	+0.003
lla_gyro	0.704/0.409	0.665/0.304	0.683/0.351	Mean	-0.017
lshoe_acc	0.704/0.409	0.691/0.367	0.703/0.407	Mean	-0.012
lshoe_gyro	0.704/0.409	0.683/0.351	0.704/0.405	Mean	-0.020
rshoe_acc	0.704/0.409	0.682/0.367	0.702/0.389	Mean	-0.020
rshoe_gyro	0.704/0.409	0.682/0.350	0.700/0.394	Mean	-0.019
back_all	0.704/0.409	0.677/0.353	0.676/0.291	CrossDiff	+0.000
right_arm_all	0.704/0.409	0.548/0.093	0.598/0.062	Mean	-0.050
left_arm_all	0.704/0.409	0.620/0.182	0.607/0.228	CrossDiff	+0.013
shoes_all	0.704/0.409	0.647/0.252	0.697/0.372	Mean	-0.050
only_back	0.704/0.409	0.590/0.037	0.584/0.035	CrossDiff	+0.006
only_right_arm	0.704/0.409	0.593/0.065	0.600/0.152	Mean	-0.008
only_left_arm	0.704/0.409	0.569/0.062	0.591/0.031	Mean	-0.022
only_shoes	0.704/0.409	0.563/0.035	0.592/0.031	Mean	-0.028

Mid-level Both Arms (ml_both_arms)
Baseline all_real: Acc=0.8897 AF1=0.5602 MAP=0.5569 AUC=0.9535 n_classes=18 | CrossDiff Acc wins=8/22 (JSON

Pattern	Baseline Acc/AF1	CrossDiff Acc/AF1	Mean-Fill Acc/AF1	Winner	Δ Acc
all_real	0.890/0.560	—	—	—	—
back_acc	0.890/0.560	0.885/0.507	0.881/0.450	CrossDiff	+0.004
back_gyro	0.890/0.560	0.871/0.498	0.880/0.438	Mean	-0.009
rua_acc	0.890/0.560	0.876/0.480	0.868/0.475	CrossDiff	+0.007
rua_gyro	0.890/0.560	0.866/0.511	0.880/0.468	Mean	-0.014
rla_acc	0.890/0.560	0.876/0.505	0.869/0.498	CrossDiff	+0.007
rla_gyro	0.890/0.560	0.880/0.500	0.869/0.503	CrossDiff	+0.011
lua_acc	0.890/0.560	0.881/0.524	0.887/0.557	Mean	-0.006
lua_gyro	0.890/0.560	0.870/0.505	0.886/0.542	Mean	-0.016
lla_acc	0.890/0.560	0.869/0.486	0.845/0.472	CrossDiff	+0.024
lla_gyro	0.890/0.560	0.879/0.495	0.886/0.551	Mean	-0.007
lshoe_acc	0.890/0.560	0.885/0.510	0.886/0.553	Mean	-0.002
lshoe_gyro	0.890/0.560	0.884/0.521	0.889/0.554	Mean	-0.005
rshoe_acc	0.890/0.560	0.886/0.517	0.880/0.542	CrossDiff	+0.006
rshoe_gyro	0.890/0.560	0.885/0.537	0.891/0.556	Mean	-0.006
back_all	0.890/0.560	0.870/0.473	0.876/0.369	Mean	-0.006
right_arm_all	0.890/0.560	0.771/0.190	0.831/0.221	Mean	-0.059
left_arm_all	0.890/0.560	0.831/0.280	0.814/0.413	CrossDiff	+0.016
shoes_all	0.890/0.560	0.853/0.320	0.878/0.533	Mean	-0.025
only_back	0.890/0.560	0.817/0.050	0.813/0.101	CrossDiff	+0.004
only_right_arm	0.890/0.560	0.822/0.085	0.822/0.256	Mean	-0.000
only_left_arm	0.890/0.560	0.810/0.065	0.829/0.113	Mean	-0.018
only_shoes	0.890/0.560	0.719/0.058	0.819/0.050	Mean	-0.099

Device-level & only-* across tracks (CrossDiff / Mean Acc, winner)

Pattern	locomotion	hl_activity	ll_left_arm	ll_left_arm_	ll_right_arm	ll_right_arm	ml_both_arms
back_all	0.81/0.72(C)	0.63/0.63(C)	0.80/0.79(C)	0.78/0.79(M)	0.77/0.75(C)	0.68/0.68(C)	0.87/0.88(M)
right_arm_all	0.80/0.75(C)	0.50/0.59(M)	0.80/0.78(C)	0.79/0.78(C)	0.55/0.62(M)	0.55/0.60(M)	0.77/0.83(M)
left_arm_all	0.79/0.77(C)	0.62/0.53(C)	0.80/0.80(M)	0.78/0.79(M)	0.70/0.75(M)	0.62/0.61(C)	0.83/0.81(C)
shoes_all	0.78/0.84(M)	0.54/0.60(M)	0.80/0.79(C)	0.79/0.78(C)	0.74/0.78(M)	0.65/0.70(M)	0.85/0.88(M)
only_back	0.49/0.55(M)	0.24/0.28(M)	0.80/0.80(=)	0.79/0.79(=)	0.62/0.61(C)	0.59/0.58(C)	0.82/0.81(C)
only_right_arm	0.42/0.43(M)	0.37/0.34(C)	0.80/0.80(=)	0.79/0.79(M)	0.66/0.72(M)	0.59/0.60(M)	0.82/0.82(M)
only_left_arm	0.50/0.38(C)	0.22/0.29(M)	0.80/0.78(C)	0.79/0.79(C)	0.62/0.61(C)	0.57/0.59(M)	0.81/0.83(M)
only_shoes	0.45/0.38(C)	0.26/0.34(M)	0.80/0.80(=)	0.79/0.79(=)	0.60/0.61(M)	0.56/0.59(M)	0.72/0.82(M)

C = CrossDiff better Acc, M = Mean better Acc, = = tie.

Key verified findings

1) Complete-data baselines (all_real)
   - locomotion: 87.32% Acc / 89.16% AF1 (main trustworthy track)
   - hl_activity: 69.28% Acc / 69.28% AF1
   - ml_both_arms: 88.97% Acc / 56.02% AF1

2) CrossDiff is not globally better than Mean-Fill
   - locomotion 9/22, hl_activity 7/22, ml_both_arms 8/22 CrossDiff Acc wins
   - Helps when missing sensors are important and recoverable (often back/arms)
   - Often loses for shoes and many gyroscope channels

3) Example strong CrossDiff wins
   - locomotion back_all +0.099; back_acc +0.066; right_arm_all +0.053
   - hl_activity left_arm_all +0.095; lla_acc +0.099

4) Example strong Mean wins
   - locomotion shoes_all -0.068 for CrossDiff
   - hl_activity right_arm_all -0.092; shoes_all -0.060

5) Left-arm tracks need caution
   - Null-class roughly 80-85%; AF1 roughly 0.04-0.08
   - Many tiny Acc deltas; high Acc-win counts are less meaningful
   - Right-arm tracks have healthier AF1 and larger drops when right arm is missing

6) Coverage
   - Every label track evaluated all 14 sensors and all 4 device groups
   - 45 scenarios per track; metrics recomputed from saved JSON files